

NailHaptic: Continuous Value Control in VR through Nail Color Change Detection for Self-Haptics Interaction

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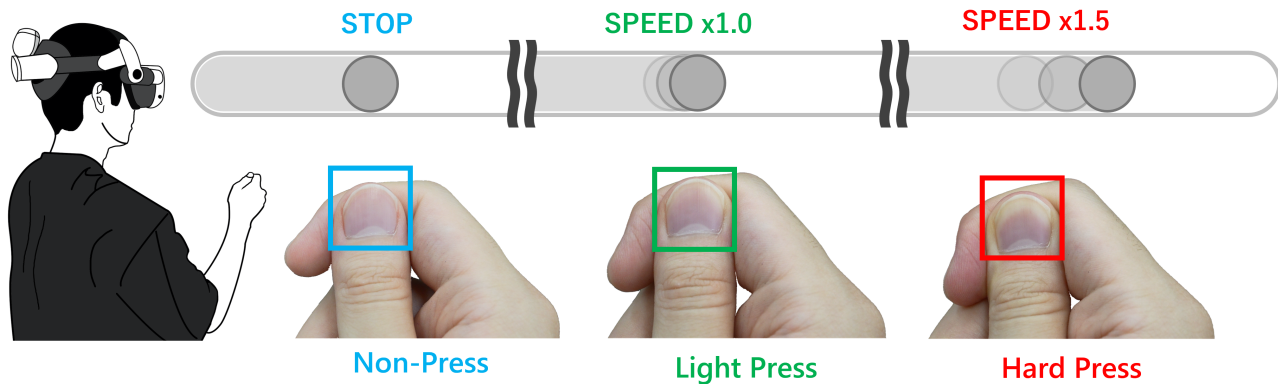


Figure 1: The NailHaptic system utilizes an RGB camera to detect pressure-induced nail color alterations during pinch gestures, facilitating continuous value UI control within VR environments. The system operates on a tiered pressure model: an absence of pressure (blue frame) ceases operation, light pressure (green frame) enables standard-speed control, and strong pressure (red frame) activates high-speed control.

Abstract

In freehand mid-air manipulation within VR environments, users' arms and hands can provide tactile cues for input operations. Conventional mid-air input lacks haptic feedback, which poses challenges for precise manipulation and continuous value control. This study proposes a novel approach that focuses on pressure variations during pinch gestures between the thumb and index finger, which are commonly used in freehand interactions. Specifically, we developed an efficient self-haptics method called NailHaptic for user interfaces (UI) with continuous value variables, by identifying the phenomenon where nail color changes according to applied pressure using RGB cameras. This system achieves the identification of three different levels of pressure applied to the nails through machine learning approaches. By integrating the machine learning model into the system, real-time pressure level determination becomes possible, demonstrating applications to intuitive UI in 3D environments.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

virtual reality, user interface

ACM Reference Format:

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1 INTRODUCTION

Self-haptics is a VR interaction technique that utilizes the user's own arm or hand as a haptic surface [1, 2, 9]. This approach complements haptic cues in freehand interaction and demonstrates higher performance [4, 5] and scalability [6, 8] compared to mid-air input. While conventional methods employ both hands, one-handed self-haptics utilizing pinch gestures offers the advantage of freeing the other hand. Examples include text input [3] and function assignment to individual finger joints [7]. However, these existing one-handed approaches are limited to discrete operations such as button touches and do not consider continuous value operations like sliders. Additionally, there are concerns about user burden caused by the placement of numerous commands.

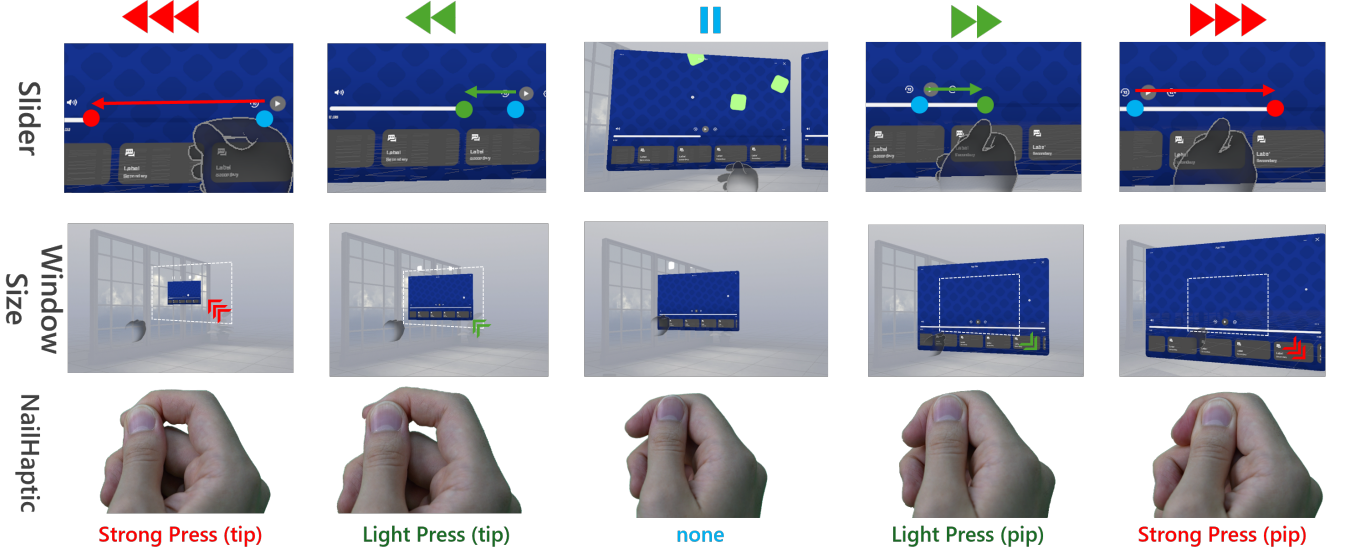


Figure 2: Applications of the NailHaptic system demonstrating continuous value control. For both slider control and window size manipulation, stronger pressure enables faster operation speeds, with pressure intensity allowing fine-grained control adjustment. The contact position between thumb and index finger determines positive/negative directional input. The bottom row shows the corresponding nail pressure states during interaction.

Therefore, this study proposes "NailHaptic," a novel self-haptics method that detects changes in finger pressure during mid-air pinch gestures as nail color variations from camera footage, enabling continuous value operations according to pressure intensity (Figure 1). This method aims to address challenges such as continuous volume operations that become difficult in mid-air input due to hand tremor and detection limitations of minute finger movements. NailHaptic is characterized by its utilization of self-haptic information—the strength of pinch pressing—for continuous value input, which is expected to provide intuitive interaction.

2 IMPLEMENTATION

The NailHaptic prototype system configuration was implemented by mounting a wide-angle RGB camera (ELP USBGS200P01-L21, 1080p) on a Meta Quest3 headset. All image processing was executed on a GPU-equipped laptop PC (3070ti), and upon detecting nail changes, data was transmitted to the Quest3 via UDP. The thumb was selected as the target for color change detection due to its large surface area, which facilitates camera capture. First, MediaPipe was used to detect thumb landmarks, and the nail region was cropped to 300×300 pixels. Subsequently, every 100 ms, we used a pre-trained segmentation model for extracting nail regions and a classification model for detecting color changes in the nails to determine the load applied to the nails.

To construct the models, a dataset was collected from 10 participants. Using an FSR pressure sensor application, three pressure levels (low: *none*: <3.0N, medium: *light*: <12.0N, high: *hard*: >12.0N) were visualized for participants, and data for each label was captured. To improve model stability, images were captured for 10 minutes each at three angles with the nail positioned frontally and

tilted left and right, collecting approximately 4,500 images per participant and 45,000 images in total. The nail segmentation model was constructed using SAM (Segment Anything Model)¹ to generate nail region masks from the dataset, followed by manual removal of inappropriate masks. This model was built using transfer learning with a U-Net architecture based on EfficientNet, which was developed by Google, achieving a training accuracy of 94.14%. The nail pressure classification model was developed as a deep learning model combining EfficientNet’s convolutional neural network (CNN) as a pre-trained encoder with a custom classification head. The final constructed model achieved a training accuracy of 96.23% and validation accuracy of 94.40%.

Furthermore, as a bidirectional control function for continuous value UI, a method was implemented to identify positive and negative directions based on which position of the index finger the thumb contacts. Specifically, MediaPipe was used to obtain the distances between the thumb tip and both the index finger tip and the second joint landmark, with input directed toward the shorter distance (finger tip: negative direction, second joint: positive direction). This enabled continuous value manipulation in both UP/DOWN directions.

3 APPLICATION

We implemented a continuous value control system in a VR environment (Meta XR All-in-One SDK) using our NailHaptic technology. The system provides intuitive control via three-stage nail pressure.

For slider-based UIs (Figure 2), such as volume or parameter controls, the pressure level dictates the rate of change. Low pressure (*none*) halts all variable changes, allowing for precise value setting.

¹<https://segment-anything.com/>

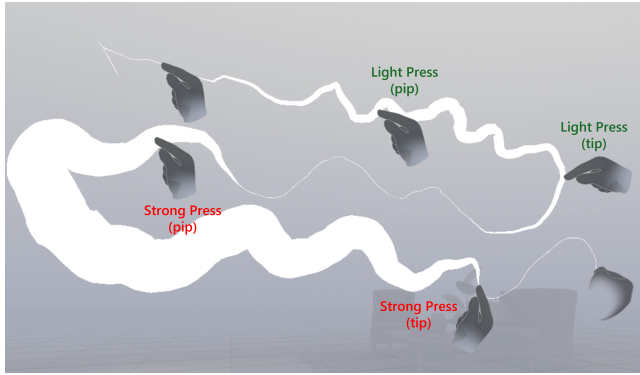


Figure 3: Applications of the NailHaptic system demonstrating a painting tool. The thickness of the brush can be changed by applying pressure when pinching.

Medium pressure (*light*) adjusts the value at 50% of a standard speed (20% of the full range per second) for fine-grained control, while high pressure (*hard*) accelerates this to 150% for efficient, large-scale changes.

We extended this interaction to other applications. For virtual display manipulation, nail pressure during a pinch gesture controls the display's scaling speed (medium: 10%/s, high: 30%/s) and its depth-positioning speed along the Z-axis (medium: 5 cm/s, high: 15 cm/s). In both cases, low pressure halts any change. Figure 2 shows an application for changing the window size.

In the application of NailHaptic to paint tools, we achieved an intuitive brush width control system in 3D space (Figure 3). Conventional freehand paint tools in 3D space require numerous command operations, including color selection, brush thickness adjustment, and brush type changes. To address this challenge, we propose a real-time brush width control system utilizing NailHaptic. The system implements a feature in which line thickness changes dynamically according to the pressure intensity applied by users during painting operations. This design achieves an intuitive user interface that aligns the natural metaphor of stronger pressure producing thicker lines with the pressing motion in pinch gestures.

4 CONCLUSION AND FUTURE WORK

This study proposes "NailHaptic," a novel method that applies self-haptics during VR freehand interaction by detecting nail color changes caused by pressure during pinch gestures using RGB cameras to enable continuous value input. Machine learning-based three-stage pressure identification suggests the possibility of precise and intuitive operation of slider UIs and virtual displays, which has been challenging with conventional air gesture approaches. The low-pressure complete stop functionality prevents unintended value fluctuations caused by hand tremor, while medium and strong pressure levels provide gradual speed control that enables efficient value adjustment. However, the proposed method faces limitations in that RGB camera-based pressure detection accuracy may deteriorate in dark environments or when nail art is applied. Future research directions include expanding the dataset and improving

machine learning models to accommodate diverse lighting conditions and nail states, conducting usability evaluations for quantitative comparison with conventional methods, and assessing fatigue levels during extended use. By addressing these challenges, NailHaptic has the potential to evolve into a more practical and versatile VR interaction method.

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